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Palaeomagnetism and Palaeosecular Variation on Lavas from Norfolk and Philip Islands, South-west Pacific Ocean

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Summary

Norfolk Island and the adjacent Philip Island in the South-west Pacific Ocean are built of basaltic lavas and tuffs that were erupted between 3·1 and 2·35 My ago during the Gauss epoch and the early part of the Matuyama epoch. Eruptions occurred at about 3·05 My ago during the Mammoth event, and between 2·8 and 2·35 My ago with one or possibly two intervals of quiescence within this period. Palaeomagnetic studies on over 200 samples from more than 80 sites, covering the whole time range of lavas found on the islands, yields an angular standard deviation of 7·5° for the directions as well as for the virtual geomagnetic poles. This value is very low compared with that expected from current models for secular variation. The deviation of the mean virtual geomagnetic pole from the Earth's rotation axis is 14°. We conclude that only a partial palaeosecular variation has been recorded in the Norfolk Island lavas because insufficient time is represented by the rocks to determine it fully, despite the fact that results on lavas of both normal and reversed polarity are included. A time span of more than 0·5 My needs to be sampled to obtain an estimate of the palaeosecular variation. The small dispersion of palaeomagnetic results from lavas in the Island of Hawaii, reported by earlier workers, may have a similar explanation to that advanced for the Norfolk Island data. If this is so then the idea that the non-dipole field under the Central Pacific has been suppressed during the Brunhes epoch needs to be re-examined.

1. Introduction

The palaeomagnetic study of a number of lava flows erupted over a period of time shows a dispersion of the measured directions of magnetization. This dispersion at least in part reflects fluctuations in the direction of the geomagnetic field, and provided that a sufficiently long period of time has been sampled an estimate of the palaeosecular variation (PSV) can be obtained from these data. Studies of the PSV for different geological times and at different localities throughout the world provides valuable information concerning the behaviour of the geomagnetic field in the past. In this paper we present the results of a palaeomagnetic study of basaltic lavas of

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Norfolk and Philip Islands in the south-west Pacific Ocean, with particular emphasis being placed on attempting to estimate the PSV during the Gauss polarity epoch and the early part of the Matuyama epoch.

2. The geomagnetic field and secular variation

The geomagnetic field is produced by a dynamo process that operates within the fluid core of the Earth (Malkus 1968). Using spherical harmonic analysis the magnetic field is separated into a dipole component and a non-dipole component. The latter is obtained by subtracting the field corresponding to the dipole from the total magnetic field (Bullard *et al.* 1950). The dipole is best represented at present by a dipole inclined at 11·4° to the rotation axis of the Earth in the meridional plane of longitude 69·8° W (Doell & Cox 1971). Observations of the geomagnetic field during historic times indicate long-period fluctuations which are collectively termed secular variation (S.V.).

The non-dipole portion of the geomagnetic field is considered to be associated with fluctuations or eddies in the convection patterns within the fluid core close to the mantle–core boundary. The non-dipole field produces broad magnetic anomalies of continental dimensions which are unevenly distributed over the Earth. These foci have a life span of a few hundred years (Creer 1962) and historical records indicate that they drift westward at a rate of about 0·2° of longitude per year. Fluctuations in the non-dipole field are thought to be the main cause of secular variation, and the periods involved are estimated to be in the order of 10^2 to 10^3 yr.

Palaeomagnetic data on rocks formed during the last few million years indicate that the average orientation of the dipole is parallel to the Earth's axis of rotation (Cox & Doell 1960; Irving 1964). The effect of the SV in the past, known as palaeosecular variation (PSV), can be estimated from the dispersion observed in palaeomagnetic directions. The precession of the dipole around the rotation axis (the dipole wobble) provides an important contribution to the PSV. The period of the precession is still very poorly known but is probably in the order of 10^4 to 10^5 years (Cox & Doell 1964; Doell & Cox 1965). Cox (1970) and Doell & Cox (1971) estimate that for the Brunhes polarity epoch the angular variance of VGPs due to dipole wobble is about 11°, whereas for earlier times Brock (1971) suggests a much smaller figure of 1½° to 2°. In addition to dipole wobble changes of intensity of the main field contribute to the PSV. To account for the palaeomagnetic observations models of the geomagnetic field have been derived by a number of workers (Cox 1962; Creer 1962; Irving & Ward 1964; Cox 1970).

In order to estimate the PSV for a particular geological epoch it is obviously necessary to measure a sufficient number of samples covering a time span that is considerably longer than the longest period of fluctuation of the geomagnetic field. Such an attempt has been made on Norfolk Island where samples from lavas that were erupted at discrete times over a period of about 0·7 My, within the Gauss and early Matuyama polarity epochs, have been measured.

3. Geology

Norfolk Island is situated at longitude 167° 57′ east and latitude 29° 02′ south on the submarine Norfolk Ridge which extends from New Zealand to New Caledonia (Fig. 1). The island is 8 km long and 5 km wide and has an area of about 35 km^2 (Fig. 1). Except for part of the south coast the island is rimmed by precipitous cliffs up to 90 m high. The average elevation of Norfolk Island is about 90 m with a roughly radial pattern of drainage developed around Mt Bates and Mt Pitt which both rise from the plateau to an altitude of about 317 m in the north-western part of the island. Philip Island lies about 6·5 km south of Norfolk Island and attains an altitude of about 274 m near its southern point (Fig. 1).

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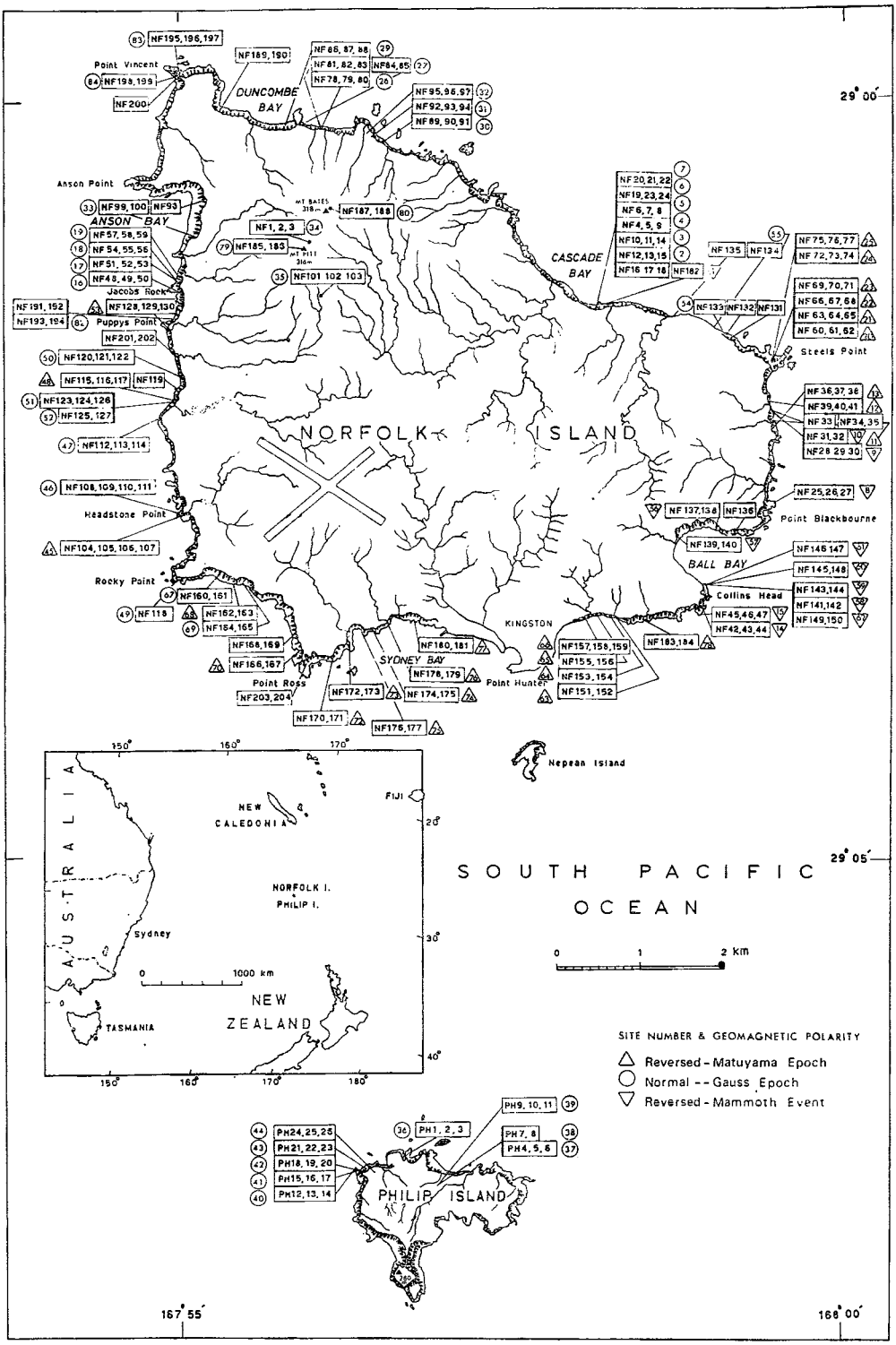


FIG. 1. Map of Norfolk Island showing its location in the south-west Pacific Ocean, and the sampling sites. Sample numbers relating to the same flow unit are enclosed in a rectangular box or in horizontally adjacent boxes. The rectangular boxes are arranged in stratigraphic order (youngest upward) where samples were collected from successive flow units in a section. Site numbers are given inside the symbol which indicates the magnetic polarity and polarity interval as given in the legend.

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In 1970 J. J. Veevers and J. G. Jones of Macquarie University and the present authors made a detailed geological study of these islands, the full results of which will be reported elsewhere. Sampling for this palaeomagnetic investigation was undertaken at that time and an additional collecting trip was made in 1971.

Norfolk Island consists mainly of basaltic lavas and tuffs as noted by earlier workers (Laing 1912; Hutton & Stephens 1956). There are few outcrops of fresh basalt inland owing to extensive weathering; however excellent exposures of unweathered basalts occur in some of the cliffs around the island. The lavas generally are flatlying or have an original dip of a few degrees. Individual flows and flow units range from 1 m to 30 m thick, and commonly show well developed columnar jointing. Up to 12 superposed lava flows, with or without interbedded tuffs, are exposed in the cliffs at any one place, and such sequences frequently can be traced laterally for several kilometres.

Potassium–argon dating of the lavas (McDougall & Aziz-ur-Rahman 1972) indicates that the volcanic activity was confined to the Late Pliocene, commencing about 3·1 My ago and ending about 2·35 My ago. The earliest lavas exposed on Norfolk Island comprise a sequence, dipping gently to the east, restricted almost entirely to the east coast from Ball Bay to Steels Point (Fig. 1). This sequence was erupted over a short period of time during the Mammoth reversed polarity event in the Gauss normal polarity epoch about 3·05 My ago (Fig. 2). We have no definite evidence as to the point of eruption of these lavas as they are surrounded inland, and overlain by younger lavas and because of the poor outcrop away from the coast. Following a hiatus of about 0·3 My eruption occurred from the general region underlying Mt Pitt and Mt Bates in the north-western part of Norfolk Island. During this second period of volcanism, which took place between about 2·75 and 2·35 My ago in the latter half of the Gauss normal polarity epoch and the early part of the Matuyama reversed epoch (Fig. 2), much of Norfolk Island, as preserved subaerially today, was built. At least one and possibly two periods of quiescence of duration about 0·1 My are recognized during this second volcanic episode.

Philip Island consists mainly of several somewhat denuded volcanic tuff cones, but at least six basaltic lavas were erupted in the final stages of activity. The lavas have K–Ar ages between 2·6 and 2·8 My, so that Philip Island was producing lavas in the Gauss normal polarity epoch at the same time as the earlier part of the second volcanic episode of Norfolk Island (Fig. 2).

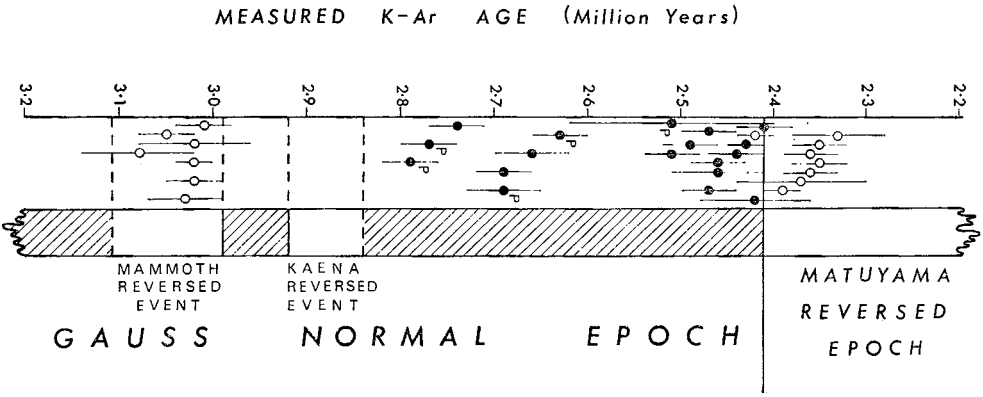


FIG. 2. Plot of K–Ar age versus magnetic polarity for the Norfolk Island and Philip Island rocks. Data, including the time scale, are from McDougall & Aziz-ur-Rahman (1972). Open circles, reversed polarity; filled circles, normal polarity. Error bars on ages are one standard deviation. Points labelled P are results on rocks from Philip Island; the remainder are results from Norfolk Island rocks.

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4. Experimental methods

A total of 230 oriented hand samples were collected from 87 different sites on Norfolk and Philip Islands. On the first excursion an average of three samples were obtained at each of the sites, which were mainly in selected flow sequences. A Sun compass was used in the orientation of all the samples. During the second excursion sampling was carried out along the south coast of Norfolk Island, as collections were not made here previously, and from a number of localities which seemed important in the light of the results already available from the first collection. This time, owing to overcast weather, orientation was done mainly with a prismatic compass, and generally only two samples were collected from each site. On both visits field polarity measurements made with a portable fluxgate magnetometer (Doell & Cox 1967) were found to be of considerable value in elucidating the structure and volcanic stratigraphy

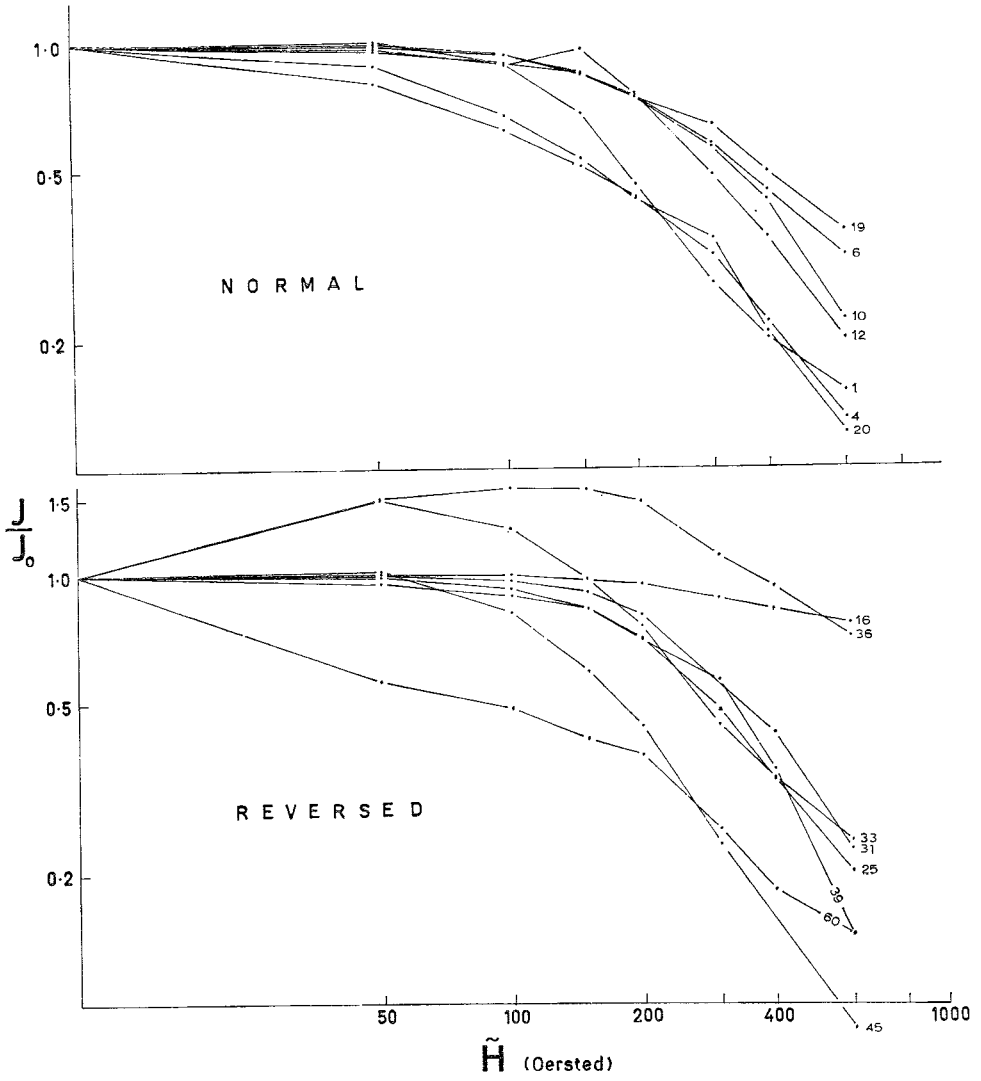


FIG. 3. Demagnetization curves for typical normal and reversed polarity samples from Norfolk Island. The ordinate is the ratio between intensity of magnetization after treatment in alternating magnetic field H and the intensity of NRM. The site number is marked on each curve.

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of Norfolk Island, and in helping to decide which sequences were those most appropriate for sampling.

At least two $2{\cdot}2 \times 2{\cdot}2$ cm cylindrical specimens were cored from each sample. Following measurement of NRM one specimen from each site was partially demagnetized in alternating fields of 50, 100, 150, 200, 300, 400 and 600 Oe (peak value) to determine the optimum field for magnetic cleaning. The partial demagnetization was done in a biaxial tumbling apparatus equipped with a programmable demagnetizing cycle and a device for decreasing the field mechanically (McElhinny 1966; Evans 1969). In most cases a field of 200 Oe (peak value) was found to be appropriate. All the measurements were made on a PAR-SM1 spinner magnetometer and the computations were carried out on an IBM 360/50 computer.

5. Results

Demagnetization curves for representative pilot specimens are shown in Fig. 3. The shapes of the curves for normal and reversed samples is consistent with the expectation that the primary magnetization acquired by the samples was TRM (Doell & Cox 1963).

Table 1

Site-mean palaeomagnetic results
(*Mean site location is at* $29{\cdot}1°$ *S*, $167{\cdot}9°$ *E*)

Site No.	*N*	*R*	Decl.	Incl.	Sample No.
1	4	3·9939	191·2	42·6	NF 16, 17, 18, 182
2	2	1·9999	20·9	−45·0	NF 12, 13*, 15
3	3	2·9933	10·9	−32·8	NF 10, 11, 14
4	3	2·9945	11·5	−36·8	NF 4, 5, 9
5	3	2·9931	8·8	−36·1	NF 6, 7, 8
6	3	2·9993	6·2	−38·9	NF 19, 23, 24
7	3	2·9848	10·6	−38·6	NF 20, 21, 22
8	3	2·9984	194·0	42·6	NF 25, 26, 27
9	3	2·9992	190·2	41·5	NF 28, 29, 30
10	2	1·9970	198·1	42·5	NF 31, 32
11	3	2·9986	196·6	40·3	NF 33, 34, 35
12	2	1·9650	195·9	27·0	NF 39, 40, 41*
13	2	1·9980	203·8	22·6	NF 36, 37*, 38
14	3	2·9989	193·7	42·2	NF 42, 43, 44
15	3	2·9911	185·3	41·9	NF 45, 46, 47
16	3	2·9948	4·3	−38·1	NF 48, 49, 50
17	2	1·9859	9·2	−43·5	NF 51*, 52, 53
18	3	2·9931	4·3	−39·2	NF 54, 55, 56
19	3	2·9961	8·0	−38·9	NF 57, 58, 59
20	3	2·9966	201·8	24·8	NF 60, 61, 62
21	2	1·9992	200·0	32·4	NF 63*, 64, 65
22	3	2·9958	202·1	32·5	NF 66, 67, 68
23	3	2·9926	197·9	33·8	NF 69, 70, 71
24	3	2·9884	191·6	40·0	NF 72, 73, 74
25	2	1·9979	196·6	40·6	NF 75*, 76, 77
26	2	1·9943	20·7	−33·3	NF 78*, 79, 80
27	2	1·9799	6·4	−37·9	NF 81*, 82, 83
28	2	1·9769	9·9	−31·2	NF 84, 85
29	3	2·9985	4·3	−31·7	NF 86, 87, 88
30	3	2·9991	10·9	−45·3	NF 89, 90, 91
31	3	2·9859	23·9	−36·0	NF 92, 93, 94

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Table 1 (*continued*)

32	2	1·9946	18·0	−43·0	NF 95, 96*, 97
33	3	2·9947	12·7	−45·4	NF 98, 99, 100
34	3	2·9928	12·7	−36·4	NF 1, 2, 3
35	2	1·9953	10·7	−33·0	NF 101, 102, 103*
36	3	2·9905	19·0	−53·9	PH 1, 2, 3
37	3	2·9874	34·7	−47·7	PH 4, 5, 6
38	2	1·9939	13·7	−32·2	PH 7, 8
39	2	1·9968	7·8	−47·0	PH 9, 10, 11*
40	3	2·9977	10·6	−46·8	PH 12, 13, 14
41	3	2·9849	6·5	−43·4	PH 15, 16, 17
42	3	2·9984	1·9	−54·0	PH 18, 19, 20
43	3	2·9943	12·3	−45·2	PH 21, 22, 23
44	3	2·9981	13·8	−47·8	PH 24, 25, 26
45	4	3·9835	197·7	37·5	NF 104, 105, 106, 107
46	4	3·9930	11·5	−40·6	NF 108, 109, 110, 111
47	3	2·9957	7·6	−46·3	NF 112, 113, 114
48	4	3·9844	206·6	42·0	NF 115, 116, 117, 119
50	3	2·9929	9·1	−49·4	NF 120, 121, 122
51	2	1·9937	2·4	−37·4	NF 123*, 124, 126
52	2	1·9989	13·3	−42·4	NF 125, 127
53	3	2·9790	221·7	42·1	NF 128, 129*, 130, 191
54	3	2·9903	20·0	−31·4	NF 131, 132, 133
55	2	1·9973	10·6	−49·7	NF 134, 135
56	2	1·9787	185·9	50·4	NF 136, 137, 138*
57	2	1·9932	196·1	46·2	NF 139, 140
58	2	1·9768	195·5	37·0	NF 141, 142
59	2	1·9887	189·3	34·4	NF 143, 144
60	2	1·9715	195·2	32·6	NF 145, 148
61	2	1·9905	201·9	35·4	NF 146, 147
63	2	1·9870	204·0	43·3	NF 151, 152
64	2	1·9948	206·9	35·6	NF 153, 154
65	2	1·9749	213·5	35·5	NF 155, 156
66	2	1·9916	203·3	31·3	NF 157, 158*, 159
67	2	1·9966	14·5	−45·4	NF 160, 161
68	2	2·0000	199·4	37·8	NF 162, 163
69	2	1·9984	9·7	−39·9	NF 164, 165
70	2	1·9988	203·8	36·3	NF 166, 167
72	2	1·9999	204·8	36·2	NF 170, 171
73	2	1·9975	201·0	39·0	NF 172, 173
74	2	1·9946	197·0	46·9	NF 174, 175
75	2	1·9983	205·9	43·1	NF 176, 177
76	2	1·9974	203·1	32·7	NF 178, 179
77	2	1·9827	202·2	29·7	NF 180, 181
78	2	1·9970	205·9	35·3	NF 183, 184
79	2	1·9799	2·7	−32·2	NF 185, 186
80	2	1·9986	13·2	−38·2	NF 187, 188
82	2	1·9943	8·2	−39·6	NF 193, 194
83	3	2·9865	11·4	−47·9	NF 195, 196, 197
84	2	1·9972	8·4	−40·4	NF 198, 199
Mean	202	199·2742	14·8	−40·1	

N: number of samples.
R: magnitude of vector sum of *N* unit vectors.
Note: Sites No. 49, 62, 71, 81, 85, 86 and 87 were not included in this table (see text).
Samples marked with asterisk were not included in two-tier analysis.

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Because the main aim of this study was to attempt to determine palaeosecular variation, careful scrutiny of all data from each site for internal consistency was undertaken prior to the usual two-tier analysis procedure. The criterion for rejection or retention of results from a sample was based upon the following argument. It is observed that the within-site precisions (k) were typically about 100. For this value of k there is only one chance in 20 that a direction will be observed lying further than 14° away from the mean. Departures of greater than 14° must thus be regarded as having very low probability, and therefore data from such samples were rejected. In those cases where only two samples were collected from the site, if the two directions differed from each other by more than 14° the reesults were not included in the two-tier analysis. The samples so rejected were NF 149, 150 (site 62); NF 168, 169 (site 71); NF 189, 190 (site 81) and NF 203, 204 (site 87). However, the magnetic polarity was quite clear and consistent for each of the sites, with the exception of site 81. Samples NF 201 and NF 202 from site 86 gave internally consistent directions but the site mean direction was about 130° from the present field. Results from this site were thus not included in the two-tier analysis. The intermediate direction for site 86 may record a partial reversal of the geomagnetic field or alternatively may indicate that the samples

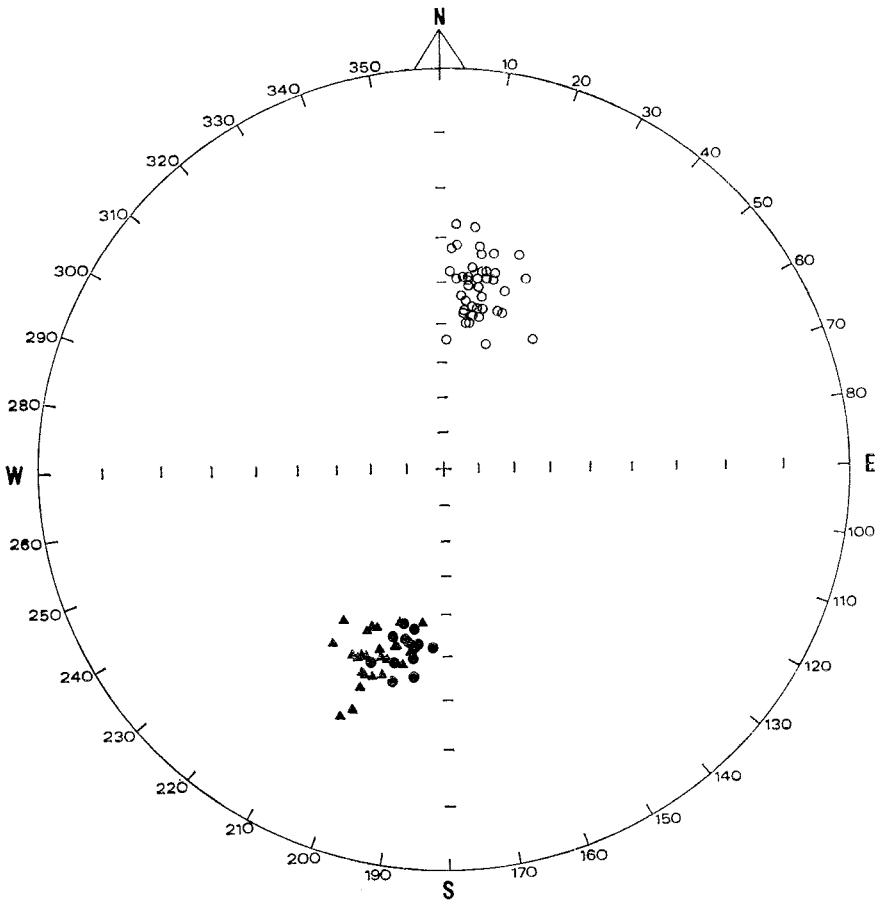


FIG. 4. Stereographic projection of site mean directions of remanent magnetization after magnetic cleaning in a field usually of 200 Oe. Open symbols indicate upward directions (Gauss normal polarity sites), closed symbols indicate downward directions (circles, Mammoth reversed polarity sites; triangles, Matuyama reversed polarity sites). Plane of projection is horizontal.

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were collected from a large slumped block. Owing to vegetational cover at this particular site we are unable to choose between the two possibilities.

Results from approximately 10 per cent of the total samples collected were rejected for two-tier analysis by the criteria given above. All acceptable site mean results are given in Table 1, where R is the resultant magnitude of N (number of samples) unit vectors (Fisher 1953). It should be noted that statistical analyses of combined site mean results using the methods of Fisher (1953) are not strictly applicable to much of the Norfolk Island data. This is because the assumption that the results were obtained from sequentially unrelated flows, the eruption of which were separated by times on average longer than the periods involved in changes of the geomagnetic field, is probably not met, as will be discussed below. Nevertheless we follow earlier workers in making the statistical analysis in order to facilitate discussion of the results.

In Fig. 1 is shown the location of the samples with sample and site numbers, together with a symbol to indicate the polarity. Thirty-four samples, whose polarity was determined, were dated by the K–Ar method, and the combined data are plotted in Fig. 2. The boundaries between polarity epochs and events are from McDougall & Aziz-ur-Rahman (1972), who have discussed the results in detail.

6. Discussion of results

(a) *General*

The direction of magnetization after alternating field treatment fall distinctly into two groups; normal and reversed, which lie approximately parallel and antiparallel to the present field directions (Fig. 4). With the possible exception of data from site 86 no intermediate directions indicating a polarity transition have been recorded. The difference between the mean directions of Gauss normal and Matuyama reversed samples is 169°. The values of α_{95} for directions for the Gauss normal and Matuyama reversed samples are 1·5° and 1·9° respectively, so that the error on the difference of 169° is about 2·4°. Thus it is apparent that the two mean directions are significantly not antiparallel.

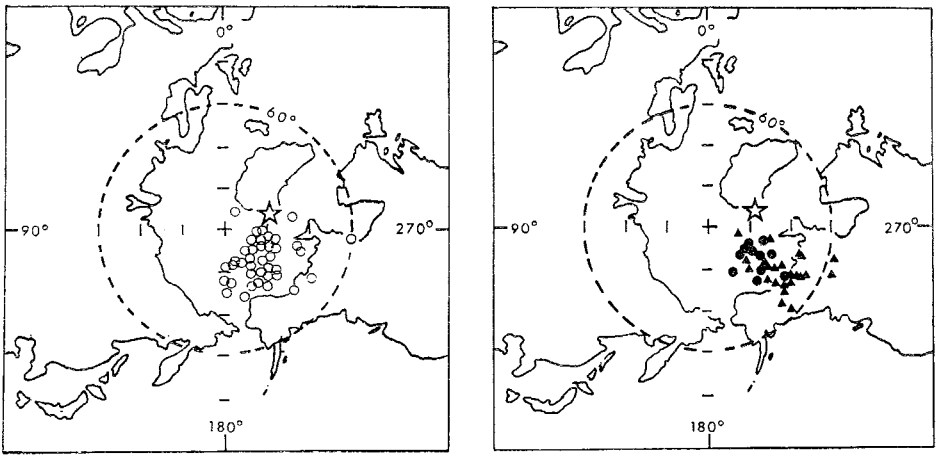


FIG. 5. Plot of virtual geomagnetic poles for all sites on north polar projections. Left-hand figure shows north VGPs for samples of the Gauss normal polarity epoch, and the right-hand figure shows south VGPs for samples of the Matuyama reversed epoch and the Mammoth reversed event. Star indicates position of present north geomagnetic pole. ○, Gauss normal; ▲, Matuyama reversed; ●, Mammoth reversed.

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Following the general practice of reversing the reversed directions the overall mean direction of 202 samples was found to be $D = 15{\cdot}1°, I = -39{\cdot}7°$ and $\alpha_{95} = 1{\cdot}2°$. The VGPs for Gauss normal, Matuyama and Mammoth reversed sites are shown in Fig. 5. The mean of all VGPs (after converting the south VGPs to north VGPs) is longitude 236·3° E, latitude 75·1° N.

(b) *Palaeosecular variation (PSV)*

In order to determine the magnitude of PSV for Norfolk and Philip Islands the data for directions and for VGPs are subjected to two-tier analysis (Watson & Irving 1957; McElhinny 1967). The two-tier analysis is carried out to separate the between-site dispersion from the within-site dispersion. In the analysis an assumption is made that the within-site dispersion is the same at all the sites; this assumption is regarded as reasonable when dealing with a sequence of basaltic lavas as in the present case. As a measure of the dispersions the between-site precision parameter β and the within-site precision parameter ω are calculated from the data. The higher the value of a precision parameter the smaller is the dispersion. The between-site precision parameter β is a measure of the magnitude of the PSV. As to whether the between-site dispersion is significant compared with the within-site dispersion is determined using the statistical F-test; if β does not differ significantly from ω then no between-site dispersion has been detected. The between-site precision parameter β also can be expressed in terms of angular dispersion or angular standard deviation:

$$\delta_\beta = \frac{81}{\sqrt{\beta}} \text{ degrees.}$$

Here the notation δ_β is identical to S referred to the VGPs and s referred to the directions as used by Cox & Doell (Cox 1970; Doell & Cox 1971). Table 2 shows the results of two-tier analysis, with β resolved into angular standard deviation and 95 per cent confidence limits, using the procedure of Cox (1969a) to obtain the latter. The two-tier analyses were made on data from the following groups of samples: Matuyama reversed, Gauss normal, Matuyama and Gauss combined, Mammoth reversed, and all samples combined.

For the Mammoth event data, referred to the mean directions, the value of $\beta = 453$ is not significantly different from ω, so that no dispersion in directions from the 12 sites sampled can be detected. The conclusion reached is that these lavas were erupted over an interval of time too short for even part of the full PSV to be recorded. The geological evidence strongly supports this conclusion. For the Mammoth reversed event lavas there are about 10 distinct and separate flows exposed on the southern side of Ball Bay and six of these were sampled for palaeomagnetic study. Potassium–argon measurements were made on three of these flows and yield ages of $3{\cdot}02 \pm 0{\cdot}03$ My, $3{\cdot}05 \pm 0{\cdot}03$ My, and $3{\cdot}01 \pm 0{\cdot}03$ My (McDougall & Aziz-ur-Rahman 1972). These ages and the lack of tuff, boles or evidence of erosion of the surface of the flows indicate that eruption of this sequence occurred over a geologically very short interval of time, probably considerably less than 40 000 yr. The other six sites in the Mammoth event lavas are more widely distributed areally (Fig. 1) and probably are from lava flows different from those sampled on the south side of Ball Bay. Ages on four of these lavas range from $3{\cdot}02 \pm 0{\cdot}06$ My to $3{\cdot}08 \pm 0{\cdot}06$ My (McDougall & Aziz-ur-Rahman 1972), again confirming that the eruptive interval was short.

Analysis of the results from the Matuyama epoch lavas indicates that β is significant, but the angular dispersion is small (Table 2). It appears that these lavas also were erupted over much too short a period for the PSV to be fully recorded. The geology is consistent with the view that eruption occurred rapidly. At Steels Point on the north-east coast of Norfolk Island six flows out of about nine exposed were sampled and K–Ar ages on four of these range from $2{\cdot}33 \pm 0{\cdot}05$ My to

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Table 2

Two-tier analyses

(a) Referred to mean directions

Samples	N	B	D_m	I_m	R_N	R_B	α_{95}	ω	β	δ_β	95% limits*
Matuyama	60	25	202·3°	+36·9°	59·3773	24·7710	1·9°	155	237	5·3°	4·7°–6·1°
Gauss	112	43	11·2°	−40·7°	110·6938	42·5742	1·5°	232	132	7·1°	6·5°–7·8°
Matuyama + Gauss	172	68	12·2°	−39·9°	169·5551	67·1386	1·8°	198	105	7·9°	7·4°–8·5°
Mammoth	30	12	193·2°	−40·9°	29·7389	11·9344	2·4°	148	453	3·8°	3·2°–4·6°
All	202	80	15·1°	−39·7°	199·2742	79·0583	1·2°	189	117	7·5°	7·0°–8·1°

(b) Referred to VGPs:

			Long. °E	Lat. °N							
Matuyama	60	25	240·6	68·2	59·4834	24·8042	1·7°	194	270	4·9°	4·3°–5·6°
Gauss	112	43	231·8	78·5	110·7693	42·6328	1·4°	223	149	6·6°	6·1°–7·3°
Matuyama + Gauss	172	68	237·1	75·2	169·6248	67·1859	1·7°	212	106	7·9°	7·4°–8·5°
Mammoth	30	12	235·7	77·0	29·7556	11·9434	2·5°	148	579	3·4°	2·9°–4·1°
All	202	80	236·3	75·1	199·3608	79·1061	1·2°	199	119	7·4°	6·9°–7·9°

*: following Cox (1969a).

N: number of samples.

B: number of sites.

R_N: length of vector sum of N unit vectors.

R_B: length of vector sum of B unit vectors.

α_{95}: 95 per cent confidence circle.

ω: within-site precision parameter.

β: between-site precision parameter.

δ_β: angular standard deviation owing to between-site scatter.

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$2{\cdot}39 \pm 0{\cdot}02$ My, virtually all agreeing to within experimental error. No breaks were recognized in the section. A total of 14 other sites, widely distributed on Norfolk Island (Fig. 1), are in Matuyama lavas; we can be confident that a minimum of four and probably at least ten of the flows are separate flows that are not represented at other sites. Four of these lavas have been dated and yield ages ranging from $2{\cdot}42 \pm 0{\cdot}02$ My to $2{\cdot}35 \pm 0{\cdot}03$ My (McDougall & Aziz-ur-Rahman 1972). These age results from the Matuyama reversed lavas suggest a period of eruption that is less than 0·10 My, and it may well have been very much less than this. Thus all the geological and other data from the Mammoth event and Matuyama epoch sequences show that each succession was erupted over a period of time insufficient to record the PSV.

The normally magnetized lavas from Norfolk and Philip Islands all belong to the last 0·4 My of the Gauss normal polarity epoch (Fig. 2). The sequences of lavas contain some interbedded tuffs and evidence of erosional breaks, and it seems likely that there were two eruptive episodes with a hiatus in the order of 0·1 My between them from 2·62 to 2·52 My (Fig. 2). However, it is clear that discrete points in time distributed over at least 0·35 My have been sampled, and yet the small between-site dispersion of the magnetic directions and VGPs is such that it is questionable whether the full PSV has been recorded. There are some redundant data included in the results because of sampling of the same flow unit at more than one locality, but it is considered that more than three-quarters of the 43 sites are from separate lava flows.

When all the data from Norfolk and Philip Islands are combined the value for the angular standard deviation is 7·5° for the directions as well as for the VGPs (Table 2). This value is still very low compared with that expected from the various models that has been advanced; for example Cox's (1970) D model gives a value of about 15° for the angular standard deviation of the VGP for this latitude.

Two alternative explanations may be advanced to account for these results. Either the PSV during the period 3·05 to 2·35 My ago was substantially less than the models for secular variation predicted for this part of the Pacific Ocean, or the amount of time represented in the sampling of the flows from Norfolk and Philip Islands is too short for the full PSV to be recorded. The second alternative appears to be the most likely explanation for two reasons. Firstly, it has already been shown that the evidence is very strong that the lavas of the Mammoth event and of the Matuyama epoch were erupted over a short time. Secondly, the deviation of the mean VGP of the Norfolk Island lavas from the axis of rotation of the Earth is 14° and thus very large. Such a deviation is not accounted for by the offset dipole effect noted by Wilson (1970a, 1971) as it is in the wrong direction and much too large. A deviation of this magnitude cannot be readily accounted for by movement of Norfolk Island subsequent to its construction. This would require that the island rotated 14° clockwise and moved southward by some 6·5° of latitude in the last 2·3 My, equivalent to a rate of movement of about 30 cm/y. Alternatively in terms of only translational movement drift in a south-easterly direction at a rate of about 60 cm/y is indicated. These rates are very large and up to an order of magnitude greater than has been estimated for this region in the late Cenozoic (Le Pichon 1968).

We therefore conclude that the angular standard deviation of the VGPs for Norfolk and Philip Islands records only a partial PSV because insufficient time is represented by the rocks exposed compared with that required for the study of the total PSV. This appears to be the case despite the fact that measurements have been made on Norfolk Island on lavas of both normal and reversed polarity that formed at different times over a period of about 0·7 My, a duration comparable to the length of the Brunhes normal polarity epoch. It has been established that for Norfolk Island both groups of reversed polarity lavas belonging to the Mammoth event and the Matuyama epoch respectively probably were erupted in periods of less than 0·1 My, and possibly substantially less. But for the Gauss normal polarity lavas an eruptive period spanning

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about 0·4 My is indicated with a possible hiatus in the activity of about 0·1 My during this time. On the basis of these data it is suggested that to study the PSV properly sampling would have to be done on rocks that have crystallized at discrete times over a period of greater than 0·5 My.

7. Comparison with studies elsewhere

Although it was concluded above that the full PSV has not been recorded in the rocks sampled from Norfolk Island it is relevant to mention briefly two recently published studies on PSV on the south-west Pacific region. Aziz-ur-Rahman (1971) studied the Newer Volcanics of Victoria in southern Australia (latitute 38° S). These lavas were erupted during the last 4·5 My and Aziz-ur-Rahman sampled from 46 sites that probably cover much of this volcanic period. He found that the angular standard deviation was 8·9° and 11·4° with respect to directions and VGPs respectively. These values are significantly lower than that predicted from most of the current models for secular variation. In contrast in a similar latitude in New Zealand, Cox (1969b) found that on the basis of 22 sites in volcanic formations of the Brunhes epoch in New Zealand that the angular standard deviation from the field of an axial dipole is 13·2° and that of the corresponding VGPs is 19·6°, substantially greater than the values expected from most models for secular variation. These two studies which have yielded strongly contrasting results emphasize the complexity of behaviour of the geomagnetic field in the past.

Of greater relevance to the present study is that a number of detailed palaeomagnetic studies have been made on lava sequences that are thought to have been erupted over very short periods of time, similar to what we have suggested for at least the reversed polarity sequences in Norfolk Island. Examples include those made on a number of flow sequences in Hawaii (Doell & Cox 1965; 1971; Doell 1969), on several stratigraphically related sequences of lavas of Tertiary age in Northern Ireland (Wilson 1970b), on a thick sequence of lavas of the Diamond Harbour Group of Banks Peninsula, New Zealand, of Late Tertiary age (Evans 1970), and on a number of lava sequences in the Aleutian Islands (Bingham & Stone 1972a). These workers concluded that because of the small dispersion of the directions and because there is generally a systematic deviation of the mean direction from the axial dipole the period of eruption was too short in the individual sequences for the full PSV to be sampled.

The careful and detailed work carried out by Doell & Cox (1963, 1965, 1971) and Doell (1969) on lava sequences from the Island of Hawaii are of particular interest in the context of the Norfolk Island study. They have examined six sequences of flows in great detail and in addition have sampled from a number of historic eruptions. All the lavas have normal polarity and were erupted during the last 0·7 My at discrete times in the Brunhes polarity epoch. Doell and Cox found that the angular variance of the results from each sequence is low, and remains low even when all the data are combined. Doell & Cox (1971) report an angular standard deviation of VGPs for the Hawaiian rocks of 11° and Doell & Cox (1965) showed that the mean direction differs by about 5° from the present axis of rotation of the Earth. They argued that each sequence was erupted over intervals of time in the order of thousands to tens of thousands of years, and for the Kau Volcanic Series, Doell (1969) and Doell & Cox (1971) suggested that eruption occurred within the last 10 000 yr. From the small dispersion of the palaeomagnetic results from Hawaii, Doell & Cox (1965, 1971) concluded that during much of the Brunhes normal polarity epoch the non-dipole field in the central Pacific area has been very small or possibly even completely absent, just as is found at the present time in this region. They interpret this to indicate that there is inhomogeneity in the lower mantle which is coupled to the fluid core of the Earth in a manner that results in suppression of the non-dipole field beneath the central Pacific, giving rise to the so-called 'Pacific window'. If this interpretation is correct it has profound geophysical implications.

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The results from Norfolk Island show very similar features to those from Hawaii in that abnormally low angular dispersion of the VGPs is found. The Norfolk Island lavas were erupted at discrete times over a total time span of about 0·7 My, similar to that sampled in the Island of Hawaii, but sampling in Norfolk Island included lavas from three separate polarity intervals in the Gauss and Matuyama epochs. However, in contrast to Hawaii, Norfolk Island at present lies in a region where the non-dipole field is large, in the order of 9000 gammas (*cf.* Doell & Cox 1971, Fig. 1; Bullard *et al.* 1950). We have given arguments above that the dispersion of the magnetic directions and VGPs for the Norfolk Island lavas is because the lavas represent an insufficiently long period of time to have recorded the full secular variation of the geomagnetic field. We therefore suggest that the low value for the apparent PSV in Hawaii may be explained similarly, and that the concept of suppression of the non-dipole field under the central Pacific during the Brunhes polarity epoch needs to be re-examined. Doell & Cox (1965, 1971) have considered the possibility that insufficient time is represented in the lavas examined in Hawaii to record the full PSV, but regarded it as an unlikely explanation for their observations. From an analysis of the available palaeomagnetic data from the Pacific Ocean region Bingham & Stone (1972b) concluded tentatively that PSV has been somewhat less for this region compared with elsewhere during the Cenozoic. However, they stressed the need for many additional results from the Pacific region, a suggestion with which we concur.

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